

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 8: Introduction to Trigonometry

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In $\triangle ABC$, right-angled at B , $AB = 24$ cm and $BC = 7$ cm. Find $\sin A$ and $\cos A$.
2. If $\cos A = \frac{4}{5}$, find $\sin A$ and $\tan A$.
3. If $\sec \theta = \frac{13}{12}$, find $\sin \theta$ and $\cot \theta$.
4. Evaluate $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.
5. Evaluate $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.
6. Evaluate $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.
7. If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, with $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
8. Prove that $(\sec A + \tan A)(1 - \sin A) = \cos A$.
9. Prove that $\frac{1 + \sec A}{\sec A} = \frac{\sin^2 A}{1 - \cos A}$.
10. Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

Answer Key

1. $AC = 25 \text{ cm}; \sin A = \frac{7}{25}, \cos A = \frac{24}{25}$

2. $\sin A = \frac{3}{5}, \tan A = \frac{3}{4}$

3. $\sin \theta = \frac{5}{13}, \cot \theta = \frac{12}{5}$

4. $\frac{3}{4} + \frac{1}{4} = 1$

5. $2 + \frac{3}{4} - \frac{3}{4} = 2$

6. $\frac{5}{4} + \frac{16}{3} - 1 = \frac{67}{12}$

7. $A = 45^\circ, B = 15^\circ$

8. $\text{LHS} = \frac{(1 + \sin A)(1 - \sin A)}{\cos A} = \frac{\cos^2 A}{\cos A} =$

9. $\text{LHS} = 1 + \cos A; \text{RHS} = \frac{(1 - \cos A)(1 + \cos A)}{1 - \cos A} = 1 + \cos A$

10. Expand: $\sin^2 A + \cos^2 A + 2 + 2 + \operatorname{cosec}^2 A + \sec^2 A = 5 + (1 + \cot^2 A) + (1 + \tan^2 A)$